

PROBLEM 9-8
Periodic Inventory Costing Procedures

Audio Shop uses a periodic inventory system. One of the most popular items carried in stock by Audio Shop is an 8-inch speaker unit. The inventory quantities, purchases, and sales of this unit for the most recent year are shown below.

	Number of Units	Cost Per Unit	Total Cost
Inventory, Jan. 1	2,700	\$30.00	\$ 81,000
First purchase (May 12)	3,540	30.60	108,324
Second purchase (July 9)	2,400	31.05	74,520
Third purchase (Oct. 4)	1,860	32.10	59,706
Fourth purchase (Dec. 18)	3,000	32.55	97,650
Goods available for sale	<u>13,500</u>		<u>\$421,200</u>
Units sold during the year	<u>9,600</u>		
Inventory, Dec. 31	<u>3,900</u>		

INSTRUCTIONS

- Using *periodic* costing procedures, compute the cost of the December 31 inventory and the cost of goods sold for the 8-inch speaker units during the year under each of the following:
 - First-in, first-out
 - Last-in, first-out
 - Average-cost
- Which of the three inventory pricing methods provides the most realistic balance sheet valuation of inventory in light of the current replacement cost of the speaker units? Does this same method also produce the most realistic measure of income in light of the costs being incurred by Audio Shop to replace the speakers when they are sold? Explain.

PROBLEM 9-9
Comparisons of Perpetual and Periodic Inventory Systems

During 1994, Playground Specialists purchased 6 BigGym redwood playground sets at the following dates and acquisition costs:

Date	Units Purchased	Unit Cost	Total Cost
Aug. 4	2	\$2,100	\$ 4,200
Sep. 23	2	2,300	4,600
Oct. 2	2	2,560	5,120
Available for sale during the year	<u>6</u>		<u>\$13,920</u>

On **September 25**, the company sold 3 of these BigGym sets to the Department of Parks and Recreation. The other 3 sets remained in inventory at December 31.

INSTRUCTIONS

- Assume that Playground Specialists uses a *perpetual inventory system*. Using each of the flow assumptions listed below, compute (a) the cost of goods sold relating to the sale of BigGym playground sets on September 25 and (b) the cost of the BigGym sets in inventory at December 31.

- Average cost
- FIFO
- LIFO

Show the number of units and the unit costs of each cost layer comprising the cost of goods sold and the ending inventory.

What amount should appear as inventory on the company's balance sheet at December 31? Explain. If you indicate an amount other than \$480,000, state which asset or liability other than inventory also would be changed in amount.

EXERCISE 9-8
Costing Inventory in a Periodic System

Herbor Company uses a *periodic* inventory system. The company's records show the beginning inventory of product no. T12 on January 1 and the purchases of this item during the current year to be as follows:

Jan. 1 Beginning Inventory.....	900 units @ \$10.00	\$ 9,000
Feb. 23 Purchase	1,200 units @ \$11.00	13,200
Apr. 20 Purchase	3,000 units @ \$11.20	33,600
May 4 Purchase	4,000 units @ \$11.80	46,400
Nov. 30 Purchase	900 units @ \$13.00	11,700
Totals	<u>10,000 units</u>	<u>\$113,900</u>

INSTRUCTIONS

A physical count indicates 1,500 units in inventory at year-end.

Determine the cost of the ending inventory, based upon each of the following methods of inventory valuation. (Remember to use *periodic* inventory costing procedures.)

- Average cost
- FIFO
- LIFO

8500 → sell

EXERCISE 9-9
Periodic Inventory Costing Methods

Pacific Plumbing uses a *periodic* inventory system. One of the company's products is a $\frac{1}{4}$ -inch brass gate valve. The company purchases these valves several times a year and makes sales of the item daily. Shown below are the inventory quantities, purchases, and sales for the year.

	Number of Units	Cost Per Unit	Total Cost
Beginning inventory (Jan. 1)	9,100	\$4.00	\$ 36,400
First purchase (Feb. 20)	20,000	4.10	82,000
Second purchase (May 10)	30,000	4.25	127,500
Third purchase (Aug. 24)	50,000	4.60	230,000
Fourth purchase (Nov. 30)	10,900	5.00	54,500
Goods available for sale	<u>120,000</u>		<u>\$530,400</u>
Units sold during the year	<u>106,000</u>		
Ending inventory (Dec. 31)	<u>14,000</u>		

INSTRUCTIONS

Compute the cost of the ending inventory of gate valves, using the following *periodic* inventory valuation methods:

- FIFO
- LIFO
- Average cost

Sold 106,000

EXERCISE 9-10
Adjusting to
Periodic LIFO

Marston Products uses a perpetual inventory system and a LIFO flow assumption. At year-end, the perpetual inventory records indicate the following units of Product RB-21 are in inventory:

Purchase Date	Quantity	Unit Cost	Total Cost
Beginning Inventory	50	\$40	\$ 2,000
June 18	300	45	13,500
Nov. 7	150	50	7,500
Total available for sale at Dec. 31	<u>500</u>		<u>\$23,000</u>

A physical count of the merchandise indicates that only 490 units of Product RB-21 are on hand. Applying **periodic LIFO** costing procedures, these 490 units would be assigned the following unit costs:

Purchase Date	Quantity	Unit Cost	Total Cost
Beginning Inventory	220	\$40	\$ 8,800
June 18	270	45	12,150
Total available for sale at Dec. 31	<u>490</u>		<u>\$20,950</u>

- INSTRUCTIONS**
- Prepare a journal entry to record the shrinkage loss of 10 units in the LIFO-based perpetual inventory records. (In recording this loss, ignore the periodic pricing procedures. Charge any loss of less than \$1,000 directly to Cost of Goods Sold.)
 - Prepare a journal entry to restate the 490 units on hand at December 31 to the unit costs indicated by periodic LIFO costing procedures.
 - Explain *why* a company using a perpetual inventory system might make a practice of restating its year-end inventory to the unit costs indicated by periodic LIFO costing procedures.

EXERCISE 9-11
Effects of Errors in Inventory Valuation

Norfleet Company prepared the following condensed income statements for two successive years:

	1994	1993
Sales	\$1,500,000	\$1,440,000
Cost of goods sold	<u>879,600</u>	<u>914,400</u>
Gross profit on sales	\$ 620,400	\$ 525,600
Operating expenses	<u>460,500</u>	<u>447,000</u>
Net Income	<u>\$ 159,900</u>	<u>\$ 78,600</u>

At the end of 1993 (right-hand column above) the inventory was understated by \$50,400, but the error was not discovered until after the accounts had been closed and financial statements prepared at the end of 1994. The balance sheets for the two years showed owner's equity of \$414,200 at the end of 1993 and \$460,400 at the end of 1994. (Norfleet is organized as a sole proprietorship and does not incur income taxes expense.)

- INSTRUCTIONS**
- Compute the corrected net income figures for 1993 and 1994.
 - Compute the gross profit amounts and the gross profit percentages for each year based upon corrected data.
 - What correction, if any, should be made in the amounts of the company's owner's equity at the end of 1993 and at the end of 1994?